

Differential Equations of the First Order and First Degree

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0$$

$$M dx + N dy = 0$$

Here M and N are functions of x and y .

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(1) Variable Separable Differential Equations

(2) Homogeneous Differential Equations

(3) Exact Differential Equations

(4) First Order Linear Differential Equations

Variables separable differential equations

A separable differential equation can be expressed as the product of a function of x and a function of y .

$$\frac{dy}{dx} = \frac{-A(x)}{B(y)}, \quad B(y) \neq 0$$

Here $A(x)$ is a function of x and $B(y)$ is a function of y .

$$A(x) + B(y) y' = 0 \quad \text{—————} \quad (1)$$

If $y = y(x)$ is a solution, it satisfies Eq (1) .

$$A(x) + B[y(x)] y'(x) = 0$$

Apply the integration operator

$$\int A(x) dx + \int B[y(x)] y'(x) dx = 0$$

$$y = y(x)$$

$$dy = y'(x) dx$$

$$\int A(x) dx + \int B(y) dy = c$$

- Differential equation generally has an infinite number of solutions.
- Such a solution is called a **general solution**.
- A corresponding initial value problem will give just **one solution**.
- A solution with no unknown constants remaining is called **specific solution**.

Example (1)

$$\frac{dy}{dx} = y^2 x^3$$

**Technique is called
separation of variables**

$$x^3 dx - \left(\frac{1}{y^2} \right) dy = 0$$

$$A(x) = x^3 \qquad B(y) = -\frac{1}{y^2}$$

$$\int x^3 dx + \int \left(-\frac{1}{y^2} \right) dy = 0$$

$$\underline{\underline{\frac{x^4}{4} + \frac{1}{y} = C}} \quad \text{Here } C \text{ is an arbitrary constant.}$$